AI ASSISTED CODING

ASSIGNMENT-10.1

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Task Description #1 – Syntax and Logic Errors

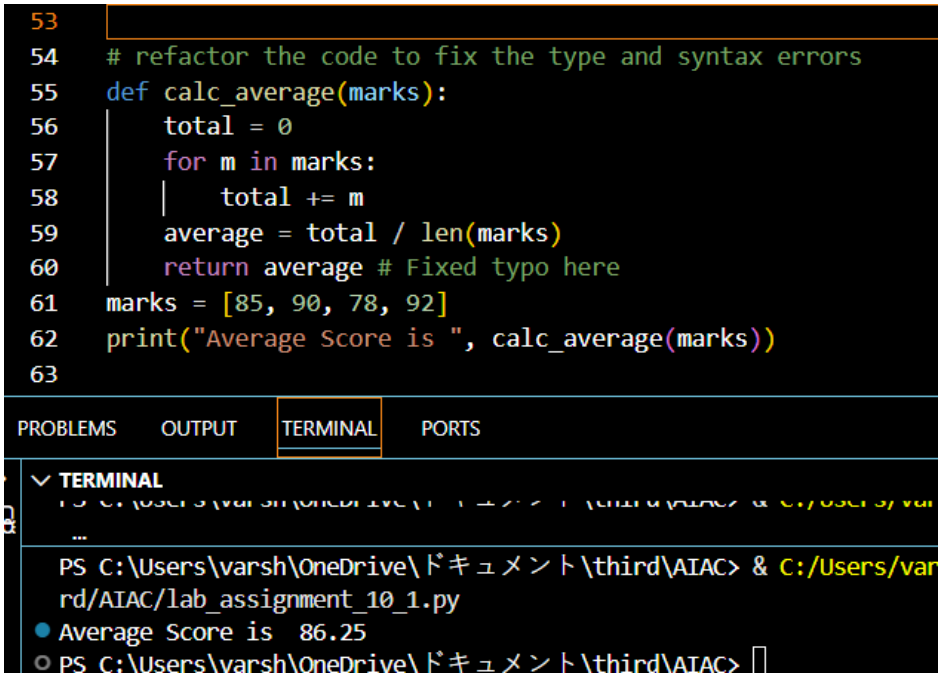
Task: Use AI to identify and fix syntax and logic errors in a faulty Python script.

Sample Input Code:

```
# Calculate average score of a student
def calc_average(marks):
total = 0
for m in marks:
total += m
average = total / len(marks)
return avrage # Typo here
marks = [85, 90, 78, 92]
print("Average Score is ", calc_average(marks)
```

Expected Output:

• Corrected and runnable Python code with explanations of the fixes.



Typo corrected – Changed return avrage to return average.

Missing bracket fixed – Added the closing) in the print() statement.

Task Description #2 – PEP 8 Compliance

Task: Use AI to refactor Python code to follow PEP 8 style guidelines.

Sample Input Code:

def area_of_rect(L,B) : return L*B

print(area_of_rect(10,20))

Expected Output:

• Well-formatted PEP 8-compliant Python code.

```
# refactor the above code to enhance readability and maintainability well commented and add type hints
def area_of_rectangle(length: float, breadth: float) -> float:
    """
    This function calculates the area of a rectangle given its length and breadth.

    Parameters:
    length (float): The length of the rectangle
    breadth (float): The breadth of the rectangle

    Returns:
    float: The area of the rectangle
    """
    return length * breadth
# Example usage:
length = 10.0
breadth = 20.0
area = area_of_rectangle(length, breadth)
print(f"The area of the rectangle with length {length} and breadth {breadth} is: {area}")
```

```
OBLEMS   OUTPUT   TERMINAL   PORTS
TERMINAL
KeyboardInterrupt
PS C:\Users\varsh\OneDrive\ドキュメント\third\AIAC> & C:/Users/varsh/AppData/Local/Programs/Python/Python313/python
rd/AIAC/lab_assignment_10_1.py
The area of the rectangle with length 10.0 and breadth 20.0 is: 200.0
PS C:\Users\varsh\OneDrive\ドキュメント\third\AIAC>
```

Task Description #3 – Readability Enhancement

Task: Use AI to make code more readable without changing its logic.

Sample Input Code:

def c(x,y):

return x*y/100

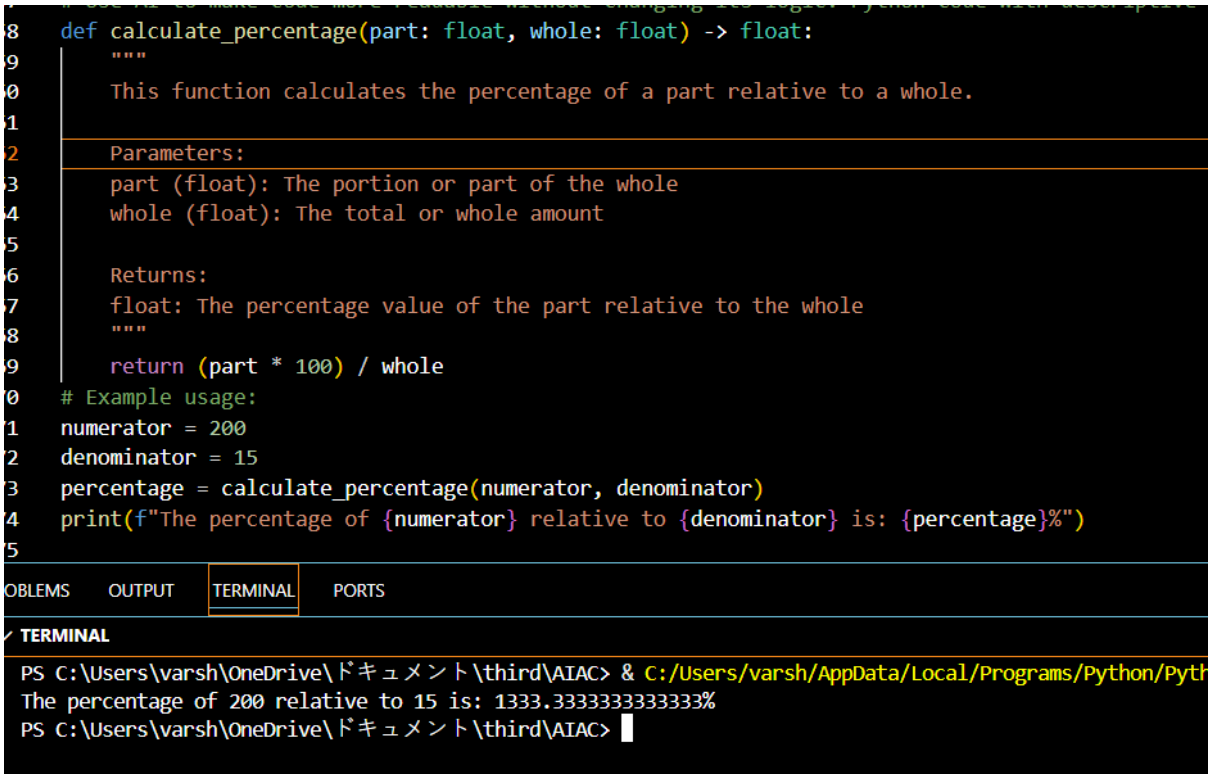
a=200

b=15

print(c(a,b))

Expected Output:

• Python code with descriptive variable names, inline comments, and clear formatting.



Task Description #4 – Refactoring for Maintainability

Task: Use AI to break repetitive or long code into reusable

functions.

Sample Input Code:

students = ["Alice", "Bob", "Charlie"]

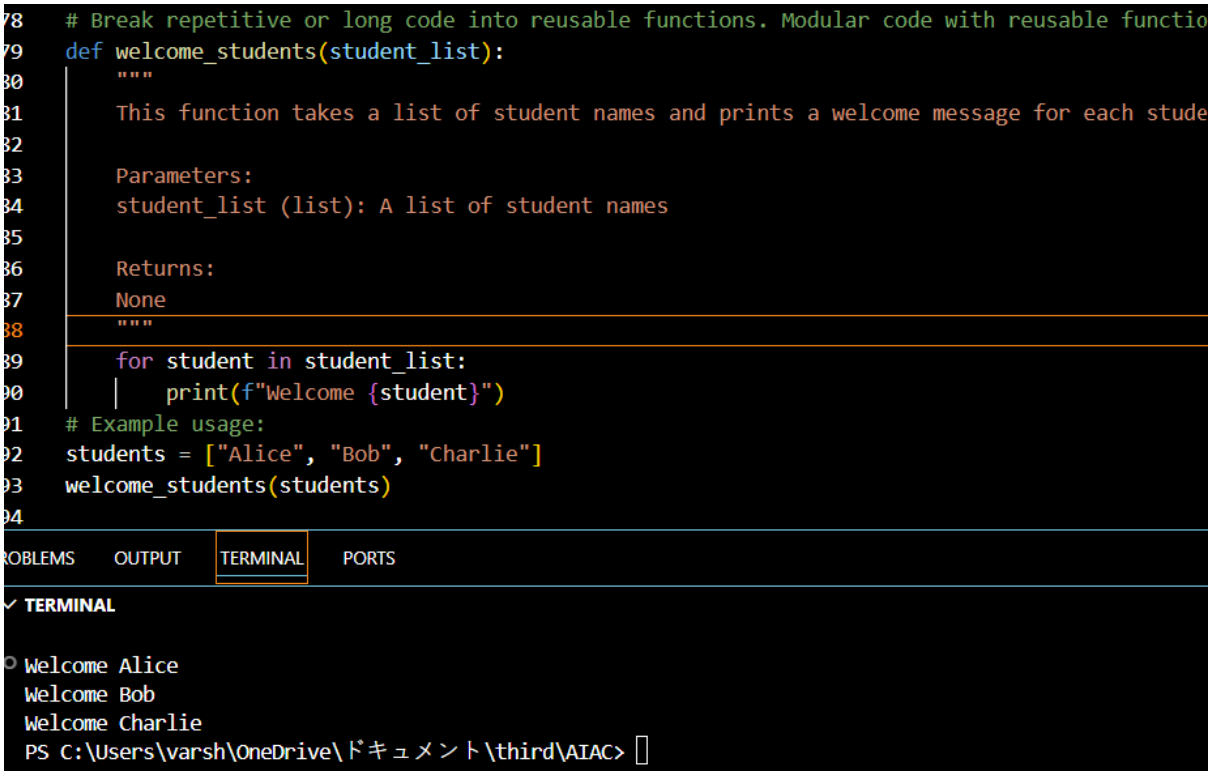
print("Welcome", students[0])

print("Welcome", students[1])

print("Welcome", students[2])

Expected Output:

• Modular code with reusable functions.



Task Description #5 – Performance Optimization

Task: Use AI to make the code run faster.

Sample Input Code:

Find squares of numbers

nums = [i for i in range(1,1000000)]

squares = []

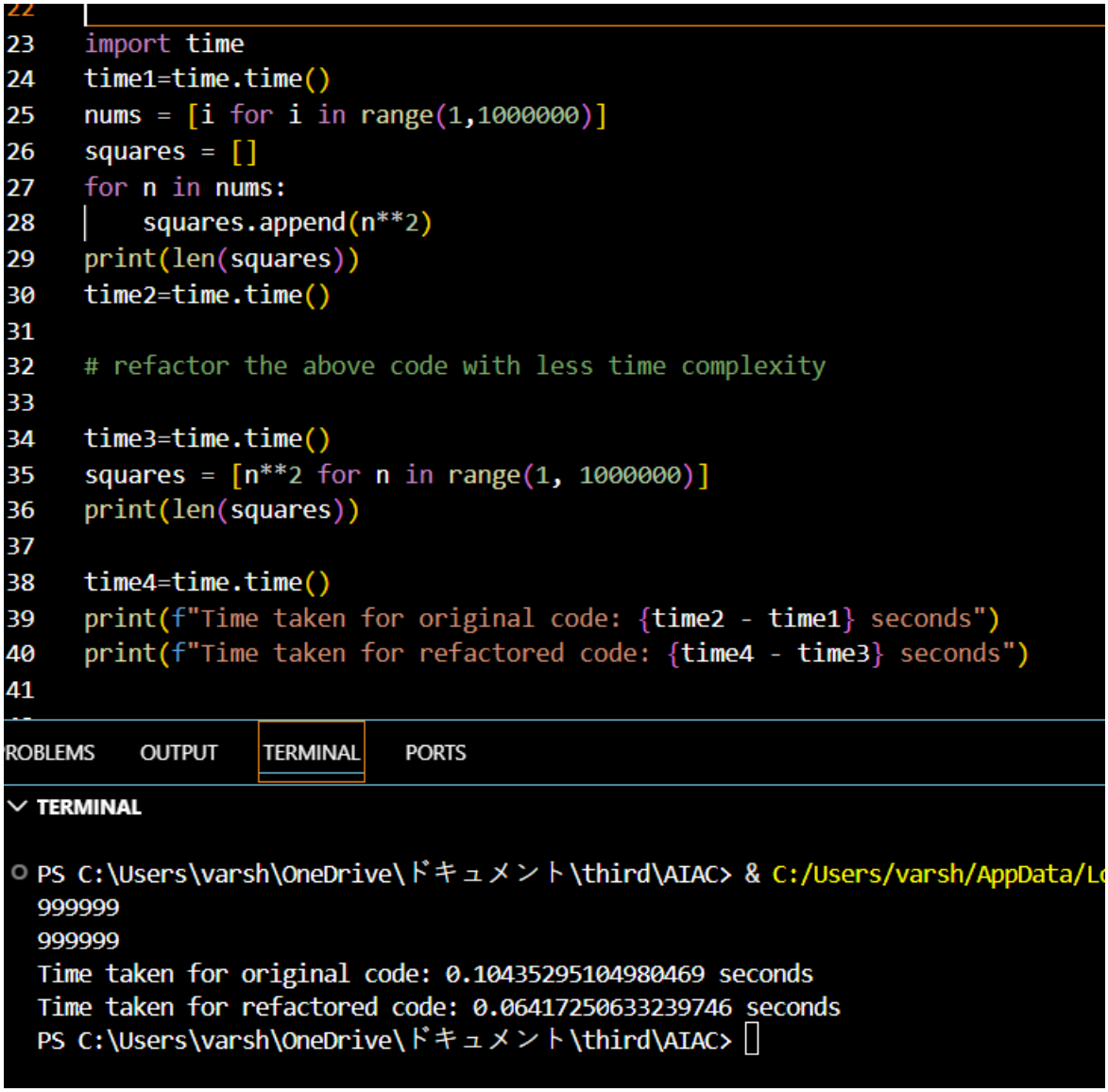
for n in nums:

squares.append(n**2)

print(len(squares))

Expected Output:

• Optimized code using list comprehensions or vectorized operations.



Task Description #6 – Complexity Reduction

Task: Use AI to simplify overly complex logic.

Sample Input Code:

def grade(score):

if score >= 90:

return "A"

else:

if score >= 80:

return "B"

else:

if score >= 70:

return "C"

else:

if score >= 60:

return "D"

else:

return "F"

Expected Output:

• Cleaner logic using elif or dictionary mapping.